Reachable, Gated, or Dead: A Dated Survey of Forty-Four Marine Biodiversity and Fisheries Data Sources

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Abstract. Building marine biodiversity or fisheries analytics tooling requires knowing which data sources are actually reachable and shaped as advertised, and that knowledge is scattered, informal, and decays. We report a dated, live-probed survey of 44 sources — occurrence aggregators, taxonomic and regional infrastructure, regional fisheries management organisations, and Indian marine research institutions — each queried on 2026-08-05 against its documented or discoverable endpoint and classified as open REST/JSON and occurrence-shaped (16), reachable but gated or bulk-download-only (21), or, for 7 Indian institutions probed separately, lacking a machine-readable API almost without exception. Three corrections to a plausible-sounding default are demonstrated with direct measurement: GBIF is not a superset of OBIS for marine fish and requires deduplication rather than a source swap; a single status-000 probe conflates a transient TLS handshake flake with a genuinely dead host; and a dead parent domain (`icar.gov.in`) does not mean a dead institution, since two of four institutes sharing it are reachable under their own `.res.in` domain. Every claim is graded *Dated*, not *Artifact*, because the object under test is infrastructure this project does not control; re-probing 9 endpoints while writing this paper, 2026-09-10, finds 2 of them already answering with a different technical signature than five weeks earlier — evidence for, not against, the paper's central claim that such a survey is worth publishing now rather than later. The sharpest finding is that no Indian marine data source runs its own machine-readable API except that this is not quite true of IndOBIS, which needs no new integration at all: it is already reachable as a node filter on the OBIS API this codebase already calls.

Keywords: biodiversity informatics, data infrastructure, OBIS, GBIF, marine fisheries data, reproducibility, API survey

1 Introduction

Building a marine species-distribution or harmful-algal-bloom model, or any analytics feature that needs occurrence, catch, or effort data broader than what one platform already fetches, starts with a question that sounds like it should

have a settled answer: which data sources are actually reachable, and in what shape? In practice, the answer is scattered across API documentation of wildly uneven quality, mailing-list folklore, a portal's own about page, and direct trial and error, and it decays — an endpoint that returned JSON last quarter can start returning an HTML redirect, a certificate error, or nothing at all, for reasons that have nothing to do with the data it holds. Sambasivan et al. document, at the level of organisational practice, that this kind of unglamorous data work is systematically undervalued relative to model work [5]; a source-reachability survey is exactly the least glamorous end of that work, which is also why it is rarely written down, and almost never written down with a date attached.

This paper reports one such survey: 44 marine biodiversity, fisheries, and catch/effort data sources, each probed live on 2026-08-05 against its documented or discoverable API endpoint. Each is classified into one of three shapes: open REST/JSON and occurrence-shaped; reachable but gated or bulk-download-only; or dead and defunct. Each verdict is reported with the specific HTTP status, DNS or TLS failure, or record count the probe returned. The sources span the field's usual anchors (OBIS, GBIF), regional and taxonomic infrastructure (WoRMS, EMODnet Biology, ICES DATRAS), regional fisheries management organisations (IOTC, ICCAT, WCPFC), and a set of Indian marine research institutions whose accessibility motivated the survey in the first place. The paper's own thesis is reflexive: a survey of this kind is useful precisely because it goes stale, which is the argument for publishing it now, with dated probes, rather than waiting for a version that will not arrive. Section 4 demonstrates this directly rather than only asserting it: re-probing a subset of the 44 sources while writing this paper, more than a month after the original pass, already finds 2 of 9 re-checked endpoints answering with a different technical signature than they did on 2026-08-05.

1.1 Contributions

1. A **dated, three-tier survey** of 44 marine biodiversity/fisheries/catch data sources, each with a live-probed verdict, the specific technical detail behind it (status code, DNS/TLS failure mode, or record count), and the date it was true.
2. Three **corrections to a plausible-sounding default**, each demonstrated with the specific measurement rather than asserted: GBIF is not a superset of OBIS for marine fish, so unioning the two requires deduplication rather than a source swap; a single non-200 probe response does not by itself mean a source is dead, since the same class of response (status 000) meant a transient TLS handshake flake for one source and a genuinely unreachable host for another; and an institution's parent-domain hostname failing does not mean the institution is unreachable, since two of four Indian institutes sharing a dead `icar.gov.in` subdomain are reachable under their own `.res.in` domain instead.
3. An explicit **evidence-grading stance**: every claim in this paper is graded *Dated* rather than *Artifact*, because the object under test is infrastructure

this project does not control, and Section 3 explains why no claim here could honestly be graded otherwise.

4. A **live re-verification pass**, performed while writing this paper, that finds the survey's own numbers already drifting in a way that argues for the paper's central claim rather than against it: an endpoint recorded dead by DNS failure on 2026-08-05 now resolves to a live host serving an unconfigured default certificate, and a portal recorded as 404 now redirects to a live page while its programmatic API remains unchanged.
5. The survey's **sharpest single finding**, isolated in Section 5: no Indian marine data source runs its own machine-readable API — except that this is not quite true of IndOBIS, which needs no new integration at all, because it is already reachable as a node filter on the OBIS API this codebase already calls.

2 Related Work

What the major aggregators' own papers describe. Grassle's founding description of the Ocean Biogeographic Information System [2] and Robertson et al.'s account of the GBIF Integrated Publishing Toolkit [4] both describe what these two aggregators are, how data enters them, and why they were built as shared infrastructure rather than per-project scraping. Costello et al. give the equivalent account for the World Register of Marine Species, the taxonomic backbone this survey lists as the right place to resolve a name before querying anything else [1]. None of the three asks, or is designed to answer, whether a given consumer can reach a given endpoint programmatically today, nor how a source outside this small, well-resourced set behaves — the question this survey's 21-source gated/bulk tier and 7-source Indian-institution tier exist to answer, since neither tier is the subject either infrastructure paper describes.

The closest prior assessment, and where this survey sits upstream of it. Vandepitte et al. assess the data quality, completeness, and fitness for use of records already inside marine biogeographic databases — duplicate records, taxonomic mismatches, geographic outliers — once a dataset has already been published into an aggregator like OBIS [6]. That is a rigorous, valuable question, and it is a different one from this survey's: Vandepitte et al. examine data already reachable and already inside the system this survey's Tier 1 describes, whereas this survey asks the question upstream of that — is a given source reachable and shaped as advertised at all — across a much longer tail of sources, most of which have never been the subject of a quality assessment because nobody has systematically probed whether they answer a query in the first place. Neither paper substitutes for the other; a source has to clear this survey's bar before Vandepitte et al.'s questions can even be asked of it.

Scope. This is a probe-based accessibility survey, not a data-quality audit and not a taxonomic or geographic coverage assessment of the sources it lists; §6

states directly that its own findings are exactly the kind of claim this paper argues decays fastest.

3 Method: what *probed* means, and why every claim here is *Dated*

Probing. Each source in Table 1, Table 2, and Table 3 was queried live against its documented or discoverable API root or a representative endpoint (an occurrence search, a statistics call, a WFS `GetCapabilities` request, or the plainest public page where no API is documented at all), recording the returned HTTP status code, DNS or TLS failure where the request did not complete, and, where the response was 200, whether the body was machine-readable (JSON, XML, or WFS) and occurrence-shaped, a generic repository schema, or plain HTML. No source was probed through an authenticated session beyond a key already provisioned in this codebase's own configuration (`backend/.env`); a 401/403 response is reported as *gated*, not retried with credentials this project does not hold. The original pass was run 2026-08-05 and recorded in `docs/data-sources.md` (split out of `CLAUDE.md` on 2026-08-28 to keep the root instruction file lean); this paper adds a second, partial pass of 9 endpoints, re-run 2026-09-10 while writing it, reported in §4.

The three-tier scheme. Tier 1 (16 sources, Table 1): open REST/JSON or equivalent (WFS, OAI-PMH) with no key required for reads, returning occurrence- or dataset-shaped data. *Tier 2* (21 sources): reachable in some form — a documentation page, a portal, a bulk-download form — but gated behind a key or account, or usable only as a file download rather than a query. *Tier 3* (7 sources): Indian marine research institutions, considered separately because they motivated the survey and because, as §5 argues, they share a specific and previously undocumented pattern that a plain Tier-1/Tier-2 split would flatten.

Why every claim here is graded Dated, not Artifact. A companion catalogue from the same research programme grades its own claims *Artifact* when a committed report file or a currently-passing regression test reproduces the number on demand, and *Dated* when the only reproduction path is a named script run again against a moving field — a day's wind, current, or SST state [3][1]. Every claim in this paper falls further along that same axis than either of that catalogue's own categories: there is no script this project owns whose re-run reproduces a source's HTTP status, because the object being measured is infrastructure entirely outside this project's control. Re-running the same request tomorrow is not *re-computation*, the way re-running a training script against a committed feature store is; it is a fresh observation of a system that can legitimately answer differently for reasons — a certificate renewal, a migrated domain, a maintenance

[1] Cross-referenced for its grading convention only; this survey has no numerical or mechanistic overlap with that catalogue's eleven instances.

window, a policy change — that have nothing to do with anything this project did. Accordingly, this paper does not use an *Artifact* grade at all: every verdict below is *Dated*, stamped with the date it was observed, and §4's re-verification pass is offered as a demonstration of exactly how much that distinction matters over as little as five weeks.

4 Results

Of 44 sources probed 2026-08-05, 16 are open, occurrence-shaped, and require no key; 21 are reachable in some form but gated or bulk-download-only; and the 7 Indian institutions in Table 3 have, with one important exception discussed below, no machine-readable API at all — everything is a PDF or HTML publication, a portal, or a manual request.

Correction 1: GBIF is not a superset of OBIS for marine fish. Unioning a second occurrence source with an existing one is often treated as a strict superset operation — more of the same kind of data, so nothing is lost by adding it. Measured directly on *Thunnus albacares*, OBIS returns 246,964 records against GBIF's 174,799 with coordinates: a large fraction of GBIF's marine records are themselves OBIS datasets republished, so a naive union double-counts, and the two sources dominate for different species groups in this codebase's own target list — GBIF for the pelagic tunas, OBIS by roughly an order of magnitude for Indian mackerel and oil sardine, the two species that matter most for Indian coastal fisheries. The union is a genuine union, not a migration, and it requires deduplication on `occurrenceID` (falling back to dataset, catalogue number, latitude/longitude, and date) precisely because neither source can be dropped for the other.

Correction 2: a single non-200 probe is not, by itself, a death certificate. Two sources across Table 1 and Table 2 returned status 000 — a connection that did not complete at all — on a single probe, and only one of them is actually dead. OTN's `/info/index.json` endpoint returned 000 on 2026-08-05, and a second request to the same host's plain HTML index succeeded; read alone, the 000 would have supported the same "dead" verdict this survey gives FishBase's ropensci mirror, whose 000 was a genuine DNS resolution failure that did not resolve on a retry. The methodological point this survey's own working notes make explicit is that a status-000 probe needs a second, different request against the same host before it is read as evidence of death rather than of a transient handshake flake — exactly the same discipline a companion catalogue from this codebase applies to a corroboration rate needing a control arm before it is read as confirmation [3].

Correction 3: a dead parent domain does not mean a dead institution. Four Indian institutes — CIFRI, CIFT, CIBA, and CIFA — share a hostname pattern under `icar.gov.in` that fails DNS/TLS for all four. Read as a single fact about

Table 1. Tier 1: open REST/JSON (or equivalent) sources, no key required for reads, occurrence- or dataset-shaped. All probed 2026-08-05 unless a re-probe date is noted; 14 rows cover 16 sources.

Source	Status	Detail
OBIS	200	`/v3/occurrence`, `/statistics`, `/checklist` all 200; `hasextensions=DNADerivedData` eDNA filter returns 44.5M records, 11,927 species, 131 datasets, 2001–2025. Already implemented (`marine_ml/sources/obis.py`).
GBIF	200	`/v1/occurrence/search`, no key, Darwin Core. Already implemented (`sources/gbif.py`); see below for why unioning it with OBIS is not a superset relation.
iNaturalist	200	`/v1/observations`, no key; citizen-science coastal bias, `quality_grade=research` mandatory; not a survey.
WoRMS	200	`/rest/AphiaRecordsByName`; taxonomic backbone for resolving a name/AphiaID before querying anything else.
Marine Regions	200	`/rest/getGazetteerRecordsByName.json`; EEZ/LME/sea-area polygons for defining a region bound instead of a hand-written bbox.
Sea Around Us	200	`/api/v1/eez`, `/eez/tonnage/taxon`; reconstructed catch by EEZ/taxon — a different table from occurrences.
ICES DATRAS	200	`/getSurveyList` (XML, SOAP-style `.asmx`); the only source in this table with true zero-catch hauls, i.e. true absences; North Atlantic/European coverage only.
EMODnet Biology	200	WFS `GetCapabilities`; `GetFeature`+bbox works without a portal scrape; European coverage.
NASA Earthdata CMR	200	`/search/collections.json` unauthenticated for search; downloads require Earthdata Login.
NOAA ERDDAP	200	Known in this codebase to flap — unloads and reloads datasets, answering a transient 404 indistinguishable from a removed dataset (applies to every ERDDAP host, OTN included).
OTN	200	Acoustic telemetry (individual tracks, not occurrences); its `/info/index.json` probe alone returned 000, a TLS handshake flake, not treated as a dead source — see the correction below.
HYCOM	200	THREDDS/OPeNDAP; environmental covariate, overlaps Copernicus, no reason to add unless a field is missing there.
Zenodo / Dryad / DataONE	200	Generic repositories; per-dataset ad-hoc schema, useful for discovering a dataset, not for a scheduled fetch.
PANGAEA	200 / 404	OAI-PMH provider (`?verb=Identify`) 200, re-probed 2026-09-10: unchanged. Its older `/es/dataportal-item/_search` endpoint is 404 — gone, use OAI-PMH instead — re-probed 2026-09-10: also unchanged.

Table 2. Tier 2: reachable but gated or bulk-download only. All probed 2026-08-05 unless a re-probe date is noted; 14 rows cover 21 sources.

Source	Status	Detail
Global Fishing Watch	401	Expected; needs the token already provisioned in `backend/.env` (`services/gfw.py`). Effort/AIS, not biology.
Movebank	401	Free account plus per-study permission from the data owner; most studies are not open.
AIMS	200 / 403	Docs 200; `api.aims.gov.au` 403 — needs an API key.
FishBase	000	`fishbase.ropensci.org` dead by DNS on 2026-08-05; re-probed 2026-09-10, its failure signature has changed — see below. `fishbase.se` serves 200 HTML only; programmatic use means the `rfishbase` data dumps, not a live API.
AquaMaps / Bio-ORACLE / MARSPEC	200	Modelled raster products downloaded as files, not queried; AquaMaps output is itself a model prediction, so using it as a label trains on another model.
IOTC / ICCAT / CCSBT	200	RFMO catch/effort, gridded aggregates behind manual download forms; ICCAT partly requires a request.
WCPFC	404	Both `/public-domain` and `/public-domain-data` 404 on 2026-08-05 — the URL moved; re-probed 2026-09-10: still 404.
FAO FishStat	200	Desktop application plus bulk zip; country-year granularity, fine for context, not for a habitat model.
NOAA Stock SMART	200 / 404	Web app 200; `/sis/api/` and `/adb/` both 404 — no documented public API, treat as bulk export.
DFO Canada	200 / 400	Federal CKAN `package_search` 200; the plain `/dataset?organization=` URL 400s.
CSIRO / NCEI / MBON / GOOS	200	Reachable only as portals.
NIWA	200 / 404	Root 200; `/coasts-and-oceans/marine-data` 404.
RLS	200	Standardised reef transects with abundance and real zeros — genuinely valuable — but ships as CSV downloads only.
VertNet	404 / 500	Portal 404, appspot API 500 on 2026-08-05; re-probed 2026-09-10, the portal's status has changed but the API has not — see below.

Table 3. Tier 3: Indian marine research institutions, no machine-readable API (one important exception). All probed 2026-08-05 unless a re-probe date is noted; 5 rows cover 7 sources.

Source	Status	Detail
CMFRI	200	`cmfri.org.in` and `eprints.cmfri.org.in` both 200; the EPrints repository has OAI-PMH and holds annual landings estimates, but document-level, not record-level.
FSI	200	Survey cruise data; publications only, data by request.
IndOBIS	200	Not a counterexample to "no machine-readable API" in the way it first appears — reachable through the existing OBIS API as node `1a3b0f1a-4474-4d73-9ee1-d28f92a83996`, no new integration needed. See §5.
CIFT / CIBA	200	`cift.res.in`, `ciba.res.in` both 200, re-probed 2026-09-10: unchanged; their `*.icar.gov.in` subdomains are 000.
CIFRI / CIFA	000 / 400	`*.icar.gov.in` subdomains 000 (DNS/TLS); `cifa.nic.in` 400. No confirmed working `.res.in` equivalent for either, unlike CIFT/CIBA.

the parent domain, this would suggest all four are simply unreachable. Two of the four, CIFT and CIBA, are reachable under their own `.res.in` domain instead (`cift.res.in`, `ciba.res.in`, both 200); the other two are not confirmed reachable under any working domain (CIFRI has no `.res.in` equivalent probed successfully here, and CIFA's `.nic.in` domain returns 400). The correction is specific and only two-for-four: an institution sharing a dead parent-domain pattern is worth a second, institution-specific probe before being written off with the other three, but that second probe does not always succeed.

What changed since 2026-08-05. Re-probing 9 of the 44 endpoints while writing this paper, 2026-09-10, finds 7 identical to the 2026-08-05 record — WCPFC's `/public-domain` still 404, PANGAEA's OAI-PMH endpoint still 200 and its old Elasticsearch endpoint still 404, `cift.res.in` and `ciba.res.in` still 200, VertNet's appspot API still 500, and the OBIS statistics call filtered to the IndOBIS node still returning exactly 79 datasets, 110,905 records, and 19,210 species over 1743–2025, byte-identical to 2026-08-05's figures. The remaining 2 have not: `fishbase.ropensci.org`, dead by DNS resolution failure on 2026-08-05, now resolves to a live host whose TLS handshake completes and serves a self-signed certificate with subject and issuer both "TRAEFIK DEFAULT CERT" (`openssl s_client`'s verify return code 18, self-signed certificate) — an ingress with no route configured for this hostname, a different failure mode from an unresolvable name, though the practical verdict (do not code against it) is unchanged. VertNet's portal, 404 on 2026-08-05, now answers 308 and redirects to a live 200 marketing page at `www.vertnet.org`, while its programmatic `appspot.com` search API is still 500 — the verdict for programmatic use (defunct) is unchanged even though the specific status code this survey's own table cites for "portal" no

longer holds. Both drifted cases were probed once more, from this session's own network access, which is a real limitation on how confidently a single re-probe can be read — discussed in §6 — but the FishBase case in particular shows a protocol-level, server-side signal (a live TLS listener answering with a specific, named default certificate) that a routing or caching difference between vantage points does not plausibly explain.

5 Discussion

The sharpest single finding in this survey is not any one dead or gated source; it is what IndOBIS's status does to the otherwise clean statement that no Indian marine data source has a machine-readable API. Read at the level of institutions, that statement holds without exception: CMFRI, FSI, CIFT, CIBA, CIFRI, and CIFA between them offer document repositories, portal pages, and manual-request forms, and not one of the six runs a queryable API of its own. IndOBIS looks, at first glance, like the seventh Indian institution and the sole counterexample — a Regional OBIS Node run by CMLRE (the Centre for Marine Living Resources and Ecology), returning 200 on its own domain (`indobis.in`) with a real dataset behind it. It is not actually a counterexample, and the reason why is the finding worth acting on: CMLRE has not built and does not run an API. IndOBIS is a *node designation inside the OBIS aggregator*, reachable by passing `?nodeid=1a3b0f1a-4474-4d73-9ee1-d28f92a83996` to the exact same `/v3/occurrence`, `/v3/dataset`, and `/v3/statistics` endpoints Table 1 already lists as open and already implemented in this codebase's `obis.py`. Filtering on that node returns 79 datasets, 110,905 records, and 19,210 species across 1743–2025 (the Actinopterygii subset alone, `taxonid=10194`, is 16,701 records across 19,210 → 3,217 species and 36 datasets), all of which a call to `obis.py` already pulls in as part of the global OBIS query it already makes.

The practical consequence is specific and easy to get wrong in the direction of doing unnecessary work: the correct response to "does this platform have Indian marine occurrence data" is not a new integration project aimed at CMLRE, because there is no CMLRE-run endpoint to integrate with — it is adding the node filter to a query already being issued, as a provenance/diagnostic facet ("how much of this region's background is Indian-sourced") rather than as a source of additional data. A survey that stopped at "IndOBIS returns 200, mark it reachable" would have filed it in the same bucket as CMFRI's eprints repository or CIFT's own domain — reachable, but requiring integration work proportional to a new source. The finding that makes this survey worth having is the one level down from reachability: *why* it is reachable, which changes the item from a backlog entry into a one-line filter on code that already ships.

6 Limitations

This survey is the work of one team, probing largely on one date (2026-08-05), against each source's documented or discoverable endpoint from this team's

own network vantage point — a different vantage point (a different country, a different institutional network, a service behind geofencing or a CDN edge) could in principle return a different status for the same source at the same moment, and neither the original pass nor this paper's re-verification pass records which network it was run from. The three-way verdict (open, gated, dead) is a deliberate simplification that several sources already strain: AIMS, DFO Canada, and NOAA Stock SMART all show a documentation page or portal succeeding while the actual data endpoint fails, which is neither cleanly "open" nor cleanly "dead," and is reported in Table 2's detail column rather than folded into a fourth category this survey does not otherwise use. The re-verification pass in §4 covers 9 of 44 sources, not all of them, chosen for being the most specific or surprising claims in the original pass rather than a random or complete sample, so 7 of 9 holding steady is evidence about those particular endpoints and should not be read as an estimate of how much of the full table has drifted.

None of this is presented as a flaw to correct in a future version, because a future, "final" version of this survey is not a coherent goal. The paper's own thesis is that this kind of survey decays on a timescale of weeks, not years, which §4's own re-verification already demonstrates for 2 of 9 endpoints checked barely five weeks after the original pass; the right response is not a more careful one-time survey but a habit of re-probing before relying on a "works" verdict months later, exactly as `docs/data-sources.md`'s own header already tells its readers to do.

7 Conclusion

Of 44 marine biodiversity, fisheries, and catch/effort data sources probed live on 2026-08-05, 16 are open and occurrence-shaped with no key required, 21 are reachable but gated or bulk-download-only, and 7 Indian institutions have, with one exception, no machine-readable API at all. That exception is the survey's sharpest finding: IndOBIS needs no new integration because it is already reachable as a node filter on the OBIS API this codebase already calls, which turns what looks like an integration backlog item into a one-line addition to an existing query. Three further corrections — that GBIF is not a superset of OBIS for marine fish, that a single status-000 probe conflates a transient flake with a genuinely dead host, and that a dead parent domain does not mean a dead institution — show that even a survey built from live probes needs a second check before a verdict is trusted, the same discipline this research programme has applied elsewhere to its own computed metrics. Every claim in this paper is graded Dated rather than Artifact, because the object under test is infrastructure outside this project's control, and a partial re-verification performed while writing this paper already finds two of nine re-checked endpoints answering with a different technical signature than they did five weeks earlier — direct evidence for, not against, the paper's central argument that a survey like this is worth publishing now, with a date attached, rather than waiting for a version that this genre of survey cannot produce.

Reproducibility

This paper's evidence is a live-probed survey of external infrastructure, not a re-runnable local computation, so its reproducibility ledger names endpoints and dates rather than committed reports or tests. The original 2026-08-05 pass is recorded in `docs/data-sources.md`, split from `CLAUDE.md` on 2026-08-28. This paper's own 2026-09-10 re-verification queried the following endpoints directly, via HTTP requests issued while writing this paper; re-issuing the same requests is the only reproduction method that applies, and may legitimately return a different answer than either date reports.

`api.obis.org/v3/statistics?nodeid=`
`1a3b0f1a-4474-4d73-9ee1-d28f92a83996`
`fishbase.ropensci.org`
`wcpfc.int/public-domain`
`ws.pangaea.de/oai/provider?verb=Identify`
`ws.pangaea.de/es/dataportal-item/_search`
`portal.vertnet.org`, `vertnet-portal.appspot.com/api/search`
`cift.res.in`, `ciba.res.in`

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